



J5

U2

C13

C11

U3

R1

C12

R2

Q1

C2

C4  
C7

C5  
C6  
C3

R3  
R4

J7

C8  
Y

J6

| No. | Quantity | Comment                                      | Designator | Footprint                    | Value | Manufacturer Part                            | Manufacturer          | Supplier Part | Supplier | Name                                         | Secondary Category |
|-----|----------|----------------------------------------------|------------|------------------------------|-------|----------------------------------------------|-----------------------|---------------|----------|----------------------------------------------|--------------------|
| 1   | 1        | 2450AT18A100E                                | ANT1       | ANT-SMD_L 3.2-W1.6           |       | 2450AT18A100E                                | JOHANSON(约翰逊)         | C89334        | LCSC     | 2450AT18A100E                                | Antennas           |
| 2   | 1        | 10uF                                         | C1         | C0603                        | 10uF  | C0603B106M007T                               | IHHEC(禾伸堂)            | C7094065      | LCSC     | 10uF                                         |                    |
| 3   | 1        | 10uF                                         | C2         | C0603                        | 10uF  | C0603B106K010T                               | IHHEC(禾伸堂)            | C2998546      | LCSC     | 10uF                                         |                    |
| 4   | 1        | 10pF                                         | C3         | C0402                        | 10pF  | C0402C0G1C100E                               | TDK                   | C19377803     | LCSC     | 10pF                                         |                    |
| 5   | 1        | 10pF                                         | C4         | C0402                        | 10pF  | C0402C0G1C100E                               | TDK                   | C19377803     | LCSC     | 10pF                                         |                    |
| 6   | 2        | 12pF                                         | C7,C8      | C0402                        | 12pF  | CC0402JRNPO7BN120                            | YAGEO(国巨)             | C1852904      | LCSC     | 12pF                                         |                    |
| 7   | 1        | 1.2pF                                        | C9         | C0402                        | 1.2pF | C0402C0G1C1R2B                               | TDK                   | C19575091     | LCSC     | 1.2pF                                        |                    |
| 8   | 1        | 1.5pF                                        | C10        | C0402                        | 1.5pF | C0402CH1C1R5C020BC                           | TDK                   | C19416497     | LCSC     | 1.5pF                                        |                    |
| 9   | 1        | 100nF                                        | C11        | C0402                        | 100nF | C0402X104K025T                               | IHHEC(禾伸堂)            | C559125       | LCSC     | 100nF                                        |                    |
| 10  | 1        | 1uF                                          | C12        | C0603                        | 1uF   | C0603X105K025T                               | IHHEC(禾伸堂)            | C559292       | LCSC     | 1uF                                          |                    |
| 11  | 1        | 10uF                                         | C13        | C0805                        | 10uF  | C2012X5R1H106KT0A0E                          | TDK                   | C25503213     | LCSC     | 10uF                                         |                    |
| 12  | 2        | 100nF                                        | C14,C18    | C0402                        | 100nF | GRM155R11C104KA88D                           | muRata(村田)            | C464413       | LCSC     | 100nF                                        |                    |
| 13  | 1        | 1uF                                          | C15        | C0402                        | 1uF   | C0402X6S105K6R3NT                            | SANYEAR(叁叶源)          | C2844323      | LCSC     | 1uF                                          |                    |
| 14  | 2        | 1uF                                          | C16,C17    | C0402                        | 1uF   | C0402X6S105K6R3NT                            | SANYEAR(叁叶源)          | C2844323      | LCSC     | 1uF                                          |                    |
| 15  | 1        | 10pF                                         | C30        | C0402                        | 10pF  | CC0402DRNPO8BN100                            | YAGEO(国巨)             | C526976       | LCSC     | 10pF                                         |                    |
| 16  | 1        | KH-TYPE-C-16P                                | J1         | USB-C-SMD_KH-TYPE-C-16P      |       | KH-TYPE-C-16P                                | kinghelm(金航标)         | C709357       | LCSC     | KH-TYPE-C-16P                                |                    |
| 17  | 1        | 6GHz                                         | J2         | RF-SMD_FR F05002-JSS103M     | 6GHz  | CONUFL001-SMD-T                              | TE Connectivity(泰科电子) | C22418213     | LCSC     | 6GHz                                         |                    |
| 18  | 2        | 2011-1X24G00SB                               | J5,J6      | HDR-TH_2011-1X24G00SB        |       | 2011-1X24G00SB                               | WingTAT(格林柏)          | C49569647     | LCSC     | 2011-1X24G00SB                               | Pin Headers        |
| 19  | 1        | SM04B-SURS-TF(LF)(SN)                        | J7         | CONN-SMD_SM04B-SURS-TF-LF-SN |       | SM04B-SURS-TF(LF)(SN)                        | JST                   | C545754       | LCSC     | SM04B-SURS-TF(LF)(SN)                        |                    |
| 20  | 1        | HX PH254-01-02-Z-L11.5 male header connector | J20        | HDR-TH_2P-P2.54-V-M          |       | HX PH254-01-02-Z-L11.5 male header connector | hanxia(韩下)            | C52016390     | LCSC     | HX PH254-01-02-Z-L11.5 male header connector |                    |

|    |   |                     |                |                                                 |       |                      |                      |           |      |                     |                                        |
|----|---|---------------------|----------------|-------------------------------------------------|-------|----------------------|----------------------|-----------|------|---------------------|----------------------------------------|
| 21 | 1 | 1.8nH               | L1             | IND-SMD_L1<br>.2-W0.6-1                         | 1.8nH | SCW1005C1<br>N8JST   | Sunltch(顺<br>磁)      | C330833   | LCSC | 1.8nH               | Inductors<br>(SMD)                     |
| 22 | 2 | YLED0603G           | LED1,LED4      | LED0603-FD                                      |       | YLED0603G            | YONGYUTAI<br>(永裕泰)   | C19273151 | LCSC | YLED0603G           |                                        |
| 23 | 1 | YLED0603R           | LED2           | LED0603-RD<br>-1                                |       | YLED0603R            | YONGYUTAI<br>(永裕泰)   | C19171390 | LCSC | YLED0603R           |                                        |
| 24 | 1 | YLED0603R           | LED3           | LED0603-FD                                      |       | YLED0603R            | YONGYUTAI<br>(永裕泰)   | C19171390 | LCSC | YLED0603R           |                                        |
| 25 | 1 | MMBT3904            | Q1             | SOT-23-3 L2<br>.9-W1.3-P1.9<br>0-LS2.4-BR       |       | MMBT3904             | CCX(晁禾)              | C51953474 | LCSC | MMBT3904            | Bipolar (BJT)                          |
| 26 | 1 | 10kΩ                | R1             | R0402                                           | 10kΩ  | CR0402JF01<br>03G    | LIZ(丽智电子<br>)        | C100572   | LCSC | 10kΩ                |                                        |
| 27 | 2 | 0Ω                  | R01,R02        | R0402                                           | 0Ω    | JR0402X20E           | Ohmite               | C2482372  | LCSC | 0Ω                  |                                        |
| 28 | 1 | 1kΩ                 | R2             | R0402                                           | 1kΩ   | CR0402FF10<br>01G    | LIZ(丽智电子<br>)        | C100195   | LCSC | 1kΩ                 |                                        |
| 29 | 2 | 10kΩ                | R3,R40         | R0402                                           | 10kΩ  | CR0402JF01<br>03G    | LIZ(丽智电子<br>)        | C100572   | LCSC | 10kΩ                |                                        |
| 30 | 2 | 4.7kΩ               | R4,R5          | R0402                                           | 4.7kΩ | CR0402JF04<br>72G    | LIZ(丽智电子<br>)        | C100658   | LCSC | 4.7kΩ               |                                        |
| 31 | 1 | 330Ω                | R6             | R0402                                           | 330Ω  | CR0402J330<br>RQ10Z  | EVER OHMS<br>(天二科技)  | C881128   | LCSC | 330Ω                |                                        |
| 32 | 1 | 10kΩ                | R7             | R0402                                           | 10kΩ  | SCR0402J10<br>K      | VO(翔胜)               | C3017497  | LCSC | 10kΩ                |                                        |
| 33 | 1 | 330Ω                | R30            | R0402                                           | 330Ω  | CR0402JF03<br>31G    | LIZ(丽智电子<br>)        | C100639   | LCSC | 330Ω                |                                        |
| 34 | 2 | 5.1kΩ               | R50,R52        | R0402                                           | 5.1kΩ | HPCR0402F<br>5K10K9  | RESI(开步睿<br>思)       | C365032   | LCSC | 5.1kΩ               |                                        |
| 35 | 2 | 1kΩ                 | R53,R54        | R0402                                           | 1kΩ   | RT0402BRD<br>071KL   | YAGEO(国巨<br>)        | C852624   | LCSC | 1kΩ                 |                                        |
| 36 | 3 | LXW-4X3X2-<br>SMD   | SW1,SW2,SW3    | SW-SMD_L3<br>.9-W3.0-LS5.<br>0                  |       | LXW-4X3X2-<br>SMD    | LXWCONN(<br>连兴旺电子)   | C53447012 | LCSC | LXW-4X3X2-<br>SMD   |                                        |
| 37 | 3 | Test-Point          | TP3,TP5,TPGND1 | Test-Point-0.<br>5mm                            |       |                      |                      |           |      |                     |                                        |
| 38 | 1 | GD32W515P<br>IQ6    | U1             | QFN-56 L7.0<br>-W7.0-P0.40-<br>TL-EP5.2         |       | GD32W515P<br>IQ6     | GigaDevice(<br>兆易创新) | C3202814  | LCSC | GD32W515P<br>IQ6    | Microcontroll<br>ers (MCU/MP<br>U/SOC) |
| 39 | 1 | AP2112K-3.3<br>TRG1 | U2             | SOT-25-5 L2<br>.9-W1.6-P0.9<br>5-LS2.8-BL       |       | AP2112K-3.3<br>TRG1  | TECH PUBLI<br>C(台舟)  | C23380830 | LCSC | AP2112K-3.3<br>TRG1 |                                        |
| 40 | 1 | CH340K              | U3             | ESOP-10 L4<br>.9-W3.9-P1.0<br>0-LS6.2-BL-E<br>P |       | CH340K               | WCH(南京沁<br>恒)        | C968586   | LCSC | CH340K              | USB<br>Converters                      |
| 41 | 1 | 40MHz               | Y1             | OSC-SMD_4<br>P-L5.0-W3.2-<br>BL_SG5032          | 40MHz | OVETGLJAN<br>F-40MHZ | TAITIEN(泰<br>艺电子)    | C295515   | LCSC | 40MHz               | Crystal<br>Oscillators                 |

| Primary Category | Category                 | Pin Count | LCSC Stock | LCSC Price | JLCPCB Stock | Add into BOM | JLCPCB Price | Convert to PCB | 3D Model Title                | Device                 | Symbol | Unique ID   | 3D Model Transform                        |
|------------------|--------------------------|-----------|------------|------------|--------------|--------------|--------------|----------------|-------------------------------|------------------------|--------|-------------|-------------------------------------------|
| RF and Wireless  | RF and Wireless:Antennas | 2         |            |            | 4872         | yes          | 0.0779       |                | ANT-SMD_L 3.2-W1.6-H1.3       | 2450AT18A100E          |        | gge4        | 125.984,62.992,0,180,0,0,0,0,0            |
|                  |                          | 2         |            |            | 0            | yes          | 0.0012       |                | C0603_L1.6-W0.8-H0.8          | C0603B106M007T         |        | gge16       | 62.992,31.496,0,0,0,0,0,0,0               |
|                  |                          | 2         |            |            | 5            | yes          | 0.0034       |                | C0603_L1.6-W0.8-H0.8          | C0603B106K010T         |        | gge17       | 62.992,31.496,0,0,0,0,0,0,0               |
|                  |                          | 2         |            |            | 0            | yes          | 0.1007       |                | C01005_L0.4-W0.2-H0.2         | C0402C0G1C100E         |        | gge14       | 15.748,7.874,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 0            | yes          | 0.1007       |                | C01005_L0.4-W0.2-H0.2         | C0402C0G1C100E         |        | gge15       | 15.748,7.874,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 571          | yes          | 0.0004       |                | C0402_L1.0-W0.5-H0.5          | CC0402JRNPO7BN120      |        | gge18,gge19 | 39.37,19.685,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 0            | yes          | 0.1007       |                | C01005_L0.4-W0.2-H0.2         | C0402C0G1C1R2B         |        | gge20       | 15.748,7.874,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 0            | yes          | 0.0419       |                | C01005_L0.4-W0.2-H0.2         | C0402CH1C1R5C020BC     |        | gge21       | 15.748,7.874,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 940          | yes          | 0.0013       |                | C0402_L1.0-W0.5-H0.5          | C0402X104K025T         |        | gge24       | 39.37,19.685,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 1724         | yes          | 0.0024       |                | C0603_L1.6-W0.8-H0.8          | C0603X105K025T         |        | gge25       | 62.992,31.496,0,0,0,0,0,0,0               |
|                  |                          | 2         |            |            | 0            | yes          |              |                | C0805_L2.0-W1.3-H1.3          | C2012X5R1H106KT0A0E    |        | gge39       | 78.74,51.181,0,0,0,0,0,0,0005,0           |
|                  |                          | 2         |            |            | 17           | yes          | 0.0005       |                | C0402_L1.0-W0.5-H0.5          | GRM155R11C104KA88D     |        | gge45,gge47 | 39.37,19.685,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 195          | yes          | 0.0008       |                | C0402_L1.0-W0.5-H0.5          | C0402X6S105K6R3NT      |        | gge43       | 39.37,19.685,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 195          | yes          | 0.0008       |                | C0402_L1.0-W0.5-H0.5          | C0402X6S105K6R3NT      |        | gge44,gge46 | 39.37,19.685,0,0,0,0,0,0,0                |
|                  |                          | 2         |            |            | 13           | yes          | 0.0005       |                | C0402_L1.0-W0.5-H0.5          | CC0402DRNPO8BN100      |        | gge13       | 39.37,19.685,0,0,0,0,0,0,0                |
|                  |                          | 20        |            |            | 312472       | yes          | 0.0114       |                | USB-C-SMD_KH-TYPE-C-16P       | KH-TYPE-C-16P          |        | gge60       | 351.9678,307.48,0,0,0,0,0,-45.276,-43.307 |
|                  |                          | 3         |            |            | 0            | yes          | 0.1527       |                | IPEX-SMD_3P-L2.6-W2.6-LS3.0   | CONUFL001-SMD-T        |        | gge5        | 122.047,118.11,0,270,0,0,0,15.984,0       |
| Connectors       | Connectors:Pin Headers   | 24        |            |            | 467          | yes          | 0.0164       |                |                               | 2011-1X24G00SB         |        | gge54,gge55 |                                           |
|                  |                          | 6         |            |            | 3869         | yes          | 0.0799       |                | CONN-SMD_SM04B-SUR S-TF-LF-SN | SM04B-SUR S-TF(LF)(SN) |        | gge52       | 212.598,106.299,0,0,0,0,0,0.001,-5.906,0  |

|                                   |                                                                   |    |  |         |        |     |        |  |                                      |                                              |  |                   |                                           |
|-----------------------------------|-------------------------------------------------------------------|----|--|---------|--------|-----|--------|--|--------------------------------------|----------------------------------------------|--|-------------------|-------------------------------------------|
|                                   |                                                                   | 2  |  |         | 39435  | yes | 0.002  |  | HDR2.54-M-LI-1X2P                    | HX PH254-01-02-Z-L11.5 male header connector |  | gge59             | 200,100,0,0,0,0,0,0,-118.11               |
| Inductors, Coils, Chokes          | Inductors, Coils, Chokes: Inductors (SMD)                         | 2  |  |         | 8773   | yes | 0.0073 |  | RES-SMD_L 1.2-W0.6-H0.6              | SCW1005C1 N8JST                              |  | gge22             | 47.244,23.622,0,0,0,0,-0.05,0,0           |
|                                   |                                                                   | 2  |  |         | 177811 | yes | 0.001  |  | LED0603_Inf                          | YLED0603G                                    |  | gge10,gge61       | 60,30,0,0,0,0,0,0,0                       |
|                                   |                                                                   | 2  |  |         | 207476 | yes | 0.001  |  | LED0603_Inf                          | YLED0603R                                    |  | gge35             | 60,30,0,0,0,0,0,0,0                       |
|                                   |                                                                   | 2  |  |         | 207476 | yes | 0.001  |  | LED0603_Inf                          | YLED0603R                                    |  | gge62             | 60,30,0,0,0,0,0,0,0                       |
| Transistors/Thyristors            | Transistors/Thyristors: Bipolar (BJT)                             | 3  |  |         | 10563  | yes | 0.001  |  | SOT-23-3P L2.9-W1.3-H1.0-LS2.4-P0.95 | MMBT3904_C51953474                           |  | gge28             | 98.425,114.173,0,180,0,0,0.0005,-0.0005,0 |
|                                   |                                                                   | 2  |  |         | 3      | yes | 0.0002 |  | R0402_L1.0-W0.5-H0.4                 | CR0402JF0103G                                |  | gge29             | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 0      | yes | 0.0219 |  | R0402_L1.0-W0.5-H0.4                 | JR0402X20E                                   |  | gge6,gge7         | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 10000  | yes | 0.0007 |  | R0402_L1.0-W0.5-H0.4                 | CR0402FF1001G                                |  | gge30             | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 3      | yes | 0.0002 |  | R0402_L1.0-W0.5-H0.4                 | CR0402JF0103G                                |  | gge12,gge31       | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 10170  | yes | 0.0003 |  | R0402_L1.0-W0.5-H0.4                 | CR0402JF0472G                                |  | gge33,gge34       | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 63     | yes | 0.0001 |  | R0402_L1.0-W0.5-H0.4                 | CR0402J330RQ10Z                              |  | gge36             | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 0      | yes | 0.0001 |  | R0402_L1.0-W0.5-H0.4                 | SCR0402J10K                                  |  | gge38             | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 279    | yes | 0.0002 |  | R0402_L1.0-W0.5-H0.4                 | CR0402JF0331G                                |  | gge11             | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 8      | yes | 0.0008 |  | R0402_L1.0-W0.5-H0.4                 | HPCR0402F5K10K9                              |  | gge26,gge27       | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 781973 | yes | 0.0037 |  | R0402_L1.0-W0.5-H0.4                 | RT0402BRD071KL                               |  | gge63,gge64       | 39.37,19.685,0,0,0,0,0,0,0                |
|                                   |                                                                   | 2  |  |         | 3141   | yes | 0.0048 |  |                                      | LXW-4X3X2-SMD                                |  | gge56,gge57,gge58 |                                           |
|                                   |                                                                   | 1  |  | 38.8823 |        | yes |        |  |                                      | Test-Point                                   |  | gge48,gge49,gge50 |                                           |
| Embedded Processors & Controllers | Embedded Processors & Controllers: Microcontrollers (MCU/MPU/SOC) | 57 |  |         | 26     | yes | 0.2045 |  | QFN-56_L7.0-W7.0-P0.40-TL-EP5.2      | GD32W515P IQ6                                |  | gge1              | 276.377,276.377,0,0,0,0,0,0,0             |
|                                   |                                                                   | 5  |  |         | 16988  | yes | 0.0162 |  | SOT-23-5_L2.9-W1.6-P0.95-LS2.8-BR    | AP2112K-3.3 TRG1_C233 80830                  |  | gge2              | 114.173,114.173,0,90,0,0,0,0,0            |

|                                         |                                                                     |    |  |  |     |     |        |  |                                                      |                      |  |       |                                        |
|-----------------------------------------|---------------------------------------------------------------------|----|--|--|-----|-----|--------|--|------------------------------------------------------|----------------------|--|-------|----------------------------------------|
| Interface                               | Interface:USB Converters                                            | 11 |  |  | 0   | yes | 0.0935 |  | ESOP-10_L4<br>.9-W3.9-H1.4<br>-P1.00-LS6.2<br>-BL-EP | CH340K               |  | gge23 | 192.913,242.<br>1255,0,0,0,0,<br>0,0,0 |
| Crystals,<br>Oscillators,<br>Resonators | Crystals,<br>Oscillators, R<br>esonators:Cr<br>ystal<br>Oscillators | 4  |  |  | 643 | yes | 0.1026 |  | SMD_5032-4<br>P_L5.0-W3.2<br>-H1.1-P2.54             | OVETGLJAN<br>F-40MHZ |  | gge8  | 196.85,125.9<br>84,0,0,0,0,0,<br>0,0   |

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| Voltage Rating (Max) | Voltage Rating | Voltage Dropout | Voltage - Supply | Viewing Angle | Vce Saturation(VCE(sat)) | Voltage - Forward(Vf) | type | Type                 | Transition frequency(ft) | Tolerance | Termination Style | Temperature Coefficient | Supplier Footprint |
|----------------------|----------------|-----------------|------------------|---------------|--------------------------|-----------------------|------|----------------------|--------------------------|-----------|-------------------|-------------------------|--------------------|
|                      |                |                 |                  |               |                          |                       |      |                      |                          |           |                   |                         | 1206               |
|                      | 6.3V           |                 |                  |               |                          |                       |      |                      |                          | ±20%      |                   | X5R                     | 0603               |
|                      | 10V            |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | X5R                     | 0603               |
|                      | 16V            |                 |                  |               |                          |                       |      |                      |                          | -         |                   | NP0 ; C0G               | -                  |
|                      | 16V            |                 |                  |               |                          |                       |      |                      |                          | -         |                   | NP0 ; C0G               | -                  |
|                      |                |                 |                  |               |                          |                       |      |                      |                          | ±5%       |                   |                         | 0402               |
|                      | 16V            |                 |                  |               |                          |                       |      |                      |                          | -         |                   | NP0 ; C0G               | -                  |
|                      | 16V            |                 |                  |               |                          |                       |      |                      |                          |           |                   | -                       | -                  |
|                      | 25V            |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | X7R                     | 0402               |
|                      | 25V            |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | X7R                     | 0603               |
|                      | 50V            |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | X5R                     | 0805               |
|                      | 16V            |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | -                       | 0402               |
|                      | 6.3V           |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | -                       | 0402               |
|                      | 6.3V           |                 |                  |               |                          |                       |      |                      |                          | ±10%      |                   | -                       | 0402               |
|                      | 25V            |                 |                  |               |                          |                       |      |                      |                          |           |                   | NP0                     | 0402               |
|                      | 30V            |                 |                  |               |                          |                       |      |                      |                          |           |                   |                         | SMD                |
|                      |                |                 |                  |               |                          |                       |      |                      |                          |           |                   |                         | SMD                |
| 600V                 |                |                 |                  |               |                          |                       |      |                      |                          |           |                   |                         | 插件,P=2.54 mm       |
|                      | 30V            |                 |                  |               |                          |                       |      |                      |                          |           |                   |                         | SMD,P=0.8mm,卧贴     |
| 1kV                  |                |                 |                  |               |                          |                       |      |                      |                          |           |                   |                         | 插件,P=2.54 mm       |
|                      |                |                 |                  |               |                          |                       |      |                      |                          | ±5%       |                   |                         | 0402               |
|                      |                |                 |                  | 120°          |                          | 2.6V~3.2V             |      |                      |                          |           |                   |                         | 0603               |
|                      |                |                 |                  | 120°          |                          | 1.8V~2.4V             |      |                      |                          |           |                   |                         | 0603               |
|                      |                |                 |                  | 120°          |                          | 1.8V~2.4V             |      |                      |                          |           |                   |                         | 0603               |
|                      |                |                 |                  |               | 300mV@50 mA,5mA          |                       | NPN  |                      | 300MHz                   |           |                   |                         | SOT-23             |
|                      |                |                 |                  |               |                          |                       |      | Thick Film Resistor  |                          | ±5%       |                   | ±200ppm/°C              | 0402               |
|                      |                |                 |                  |               |                          |                       |      | -                    |                          | -         |                   | -                       | 0402               |
|                      |                |                 |                  |               |                          |                       |      | Thick Film Resistor  |                          | ±1%       |                   | ±200ppm/°C              | 0402               |
|                      |                |                 |                  |               |                          |                       |      | Thick Film Resistors |                          | ±5%       |                   | ±200ppm/°C              | 0402               |
|                      |                |                 |                  |               |                          |                       |      | Thick Film Resistor  |                          | ±5%       |                   | ±200ppm/°C              | 0402               |
|                      |                |                 |                  |               |                          |                       |      | Thick Film Resistor  |                          | ±5%       |                   | ±100ppm/°C              | 0402               |
|                      |                |                 |                  |               |                          |                       |      | Thick Film Resistor  |                          | ±5%       |                   |                         | 0402               |

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| Frequency(C<br>enter/Band) | Frequency<br>Stability | Frequency -<br>Self<br>Resonant | Forward<br>Current | Global Net<br>Name | Features | Emitter-Base<br>Voltage(Vebo<br>) | Diode<br>Configuration | Detection<br>color | Description                                                                                                                                                                                                                                                                                 | DC Resistan<br>ce(DCR) | DC Current<br>Gain |
|----------------------------|------------------------|---------------------------------|--------------------|--------------------|----------|-----------------------------------|------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|
| 2.4GHz                     |                        |                                 |                    |                    |          |                                   |                        |                    | Antenna Type:Patch Antenna<br>Frequency(Center/Band):2.4GHz<br>Bandwidth:- Peak Gain:0.5dBi Peak<br>Gain:0.5dBi Impedance:50Ω Power:2W<br>VSWR:- Operating<br>Temperature:-40°C~+125°C Operating<br>Temperature:-40°C~+125°C Connector<br>Type:SMA Length:3.2mm Width:1.6mm<br>Height:1.3mm |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:10uF Tolerance:±20%<br>Tolerance:±20% Voltage Rating:6.3V<br>Temperature Coefficient:X5R                                                                                                                                                                                        |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:10uF Tolerance:±10%<br>Tolerance:±10% Voltage Rating:10V<br>Temperature Coefficient:X5R                                                                                                                                                                                         |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:10pF Tolerance:- Tolerance:-<br>Voltage Rating:16V Temperature<br>Coefficient:NP0 ; C0G                                                                                                                                                                                         |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:10pF Tolerance:- Tolerance:-<br>Voltage Rating:16V Temperature<br>Coefficient:NP0 ; C0G                                                                                                                                                                                         |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance: Tolerance:±5%<br>Tolerance:±5% Voltage Rated:<br>Temperature Coefficient:                                                                                                                                                                                                      |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:1.2pF Tolerance:- Tolerance:-<br>Voltage Rating:16V Temperature<br>Coefficient:NP0 ; C0G                                                                                                                                                                                        |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:1.5pF Voltage Rating:16V<br>Temperature Coefficient:-                                                                                                                                                                                                                           |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:100nF Tolerance:±10%<br>Tolerance:±10% Voltage Rating:25V<br>Temperature Coefficient:X7R                                                                                                                                                                                        |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:1uF Tolerance:±10%<br>Tolerance:±10% Voltage Rating:25V<br>Temperature Coefficient:X7R                                                                                                                                                                                          |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    |                                                                                                                                                                                                                                                                                             |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    |                                                                                                                                                                                                                                                                                             |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    |                                                                                                                                                                                                                                                                                             |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    |                                                                                                                                                                                                                                                                                             |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Capacitance:10pF Voltage Rating:25V<br>Temperature Coefficient:NP0                                                                                                                                                                                                                          |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    |                                                                                                                                                                                                                                                                                             |                        |                    |
|                            |                        |                                 |                    |                    |          |                                   |                        |                    | Connector Type:Board Side Interface:Male<br>Pin Number of Ports:1 Frequency:6GHz<br>Impedance:50Ω Operating Temperature<br>Range:-40°C~+90°C Operating<br>Temperature Range:-40°C~+90°C                                                                                                     |                        |                    |

|  |  |   |      |  |  |    |                |  |                                                                                                                                                                                                                                                                   |      |              |
|--|--|---|------|--|--|----|----------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------|
|  |  |   |      |  |  |    |                |  | Pin Structure:1x24P Pitch:2.54mm Circular Pins/Square Pins:Pin Header Mounting Type:Through Hole Number of Rows:1 Number of Pins:24P Current Rating:3A Voltage Rating (Max):600V Length of Mating Pin:6mm Insulation Height:2.5mm Length of End Connection Pin:   |      |              |
|  |  |   |      |  |  |    |                |  | Pins Structure:1x4P Pitch:0.8mm Mounting Type:Surface Mount, Right Angle Reference Series:SUR Number of Pins:4P Number of Rows:1 Number of PINs Per Row:4 Row Spacing:- Current Rating:500mA Contact Material:Copper Alloy Contact Plating:Tin Operating Temper   |      |              |
|  |  |   |      |  |  |    |                |  |                                                                                                                                                                                                                                                                   |      |              |
|  |  | - |      |  |  |    |                |  | Inductance:1.8nH Tolerance:±5% Tolerance:±5% Current Rating:950mA Current - Saturation (Isat):- DC Resistance(DCR):43mΩ Q @ Frequency:- Frequency - Self Resonant:- Ratings:-                                                                                     | 43mΩ |              |
|  |  |   | 25mA |  |  |    | Discrete Diode |  | Lamp Holder Type:- Illumination Color:Emerald Green Voltage - Forward(Vf):2.6V~3.2V Voltage - Forward(Vf):2.6V~3.2V Power(Watts):- Forward Current:25mA Wavelength - Dominant:510nm~531nm Wavelength - Dominant:510nm~531nm Peak Wavelength:- Peak Wavelength:-   |      |              |
|  |  |   | 25mA |  |  |    | Discrete Diode |  | Lamp Holder Type:- ; - Illumination Color:Red Voltage - Forward(Vf):1.8V~2.4V Voltage - Forward(Vf):1.8V~2.4V Power(Watts):- Forward Current:25mA Wavelength - Dominant:620nm~625nm Wavelength - Dominant:620nm~625nm Peak Wavelength:- Diode Configuration:Discr |      |              |
|  |  |   | 25mA |  |  |    | Discrete Diode |  | Lamp Holder Type:- ; - Illumination Color:Red Voltage - Forward(Vf):1.8V~2.4V Voltage - Forward(Vf):1.8V~2.4V Power(Watts):- Forward Current:25mA Wavelength - Dominant:620nm~625nm Wavelength - Dominant:620nm~625nm Peak Wavelength:- Diode Configuration:Discr |      |              |
|  |  |   |      |  |  | 6V |                |  | type:NPN Current - Collector(Ic):200mA Collector - Emitter Voltage VCEO:40V Pd - Power Dissipation:200mW DC Current Gain:300@10mA,1V DC Current Gain:300@10mA,1V Transition frequency(fT):300MHz Current - Collector Cutoff:100nA Vce Saturation(VCE(sat)):300m   |      | 300@10mA, 1V |

|  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                      |  |  |
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|  |  |  |  |  |  |  |  |  | Type:Thick Film Resistors Resistance:<br>Tolerance:±5% Tolerance:±5%<br>Power(Watts): Overload Voltage (Max):<br>Temperature Coefficient:±200ppm/°C<br>Temperature Coefficient:±200ppm/°C<br>Operating Temperature<br>Range:-55°C~+125°C Operating<br>Temperature Range:-55°C~+125°C |  |  |
|  |  |  |  |  |  |  |  |  | Type:- Resistance:0Ω Tolerance:-<br>Tolerance:- Power(Watts):200mW<br>Voltage-Supply(Max):- Temperature<br>Coefficient:- Temperature Coefficient:-<br>Operating Temperature:-55°C~+150°C<br>Operating Temperature:-55°C~+150°C                                                       |  |  |
|  |  |  |  |  |  |  |  |  | Type:Thick Film Resistor Resistance:1kΩ<br>Tolerance:±1% Tolerance:±1%<br>Power(Watts):62.5mW<br>Voltage-Supply(Max):50V Temperature<br>Coefficient:±200ppm/°C Temperature<br>Coefficient:±200ppm/°C Operating<br>Temperature:-55°C~+125°C Operating<br>Temperature:-55°C~+125°C     |  |  |
|  |  |  |  |  |  |  |  |  | Type:Thick Film Resistors Resistance:<br>Tolerance:±5% Tolerance:±5%<br>Power(Watts): Overload Voltage (Max):<br>Temperature Coefficient:±200ppm/°C<br>Temperature Coefficient:±200ppm/°C<br>Operating Temperature<br>Range:-55°C~+125°C Operating<br>Temperature Range:-55°C~+125°C |  |  |
|  |  |  |  |  |  |  |  |  | Type:Thick Film Resistor<br>Resistance:4.7kΩ Tolerance:±5%<br>Tolerance:±5% Power(Watts):62.5mW<br>Voltage-Supply(Max):50V Temperature<br>Coefficient:±200ppm/°C Temperature<br>Coefficient:±200ppm/°C Operating<br>Temperature:-55°C~+125°C Operating<br>Temperature:-55°C~+125°C   |  |  |
|  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                      |  |  |
|  |  |  |  |  |  |  |  |  |                                                                                                                                                                                                                                                                                      |  |  |
|  |  |  |  |  |  |  |  |  | Type:Thick Film Resistor Resistance:330Ω<br>Tolerance:±5% Tolerance:±5%<br>Power(Watts):62.5mW<br>Voltage-Supply(Max):50V Temperature<br>Coefficient:±200ppm/°C Temperature<br>Coefficient:±200ppm/°C Operating<br>Temperature:-55°C~+125°C Operating<br>Temperature:-55°C~+125°C    |  |  |
|  |  |  |  |  |  |  |  |  | Type:Thick Film Resistors<br>Resistance:5.1kΩ Tolerance:±1%<br>Tolerance:±1% Power(Watts):100mW<br>Voltage-Supply(Max):50V Temperature<br>Coefficient:±100ppm/°C Temperature<br>Coefficient:±100ppm/°C Operating<br>Temperature:-55°C~+125°C Operating<br>Temperature:-55°C~+125°C   |  |  |

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|  |        |  |  |  |                                                                                                     |  |  |  | Type:Thin Film Resistor Resistance:1kΩ<br>Tolerance:±0.1% Tolerance:±0.1%<br>Power(Watts):62.5mW<br>Voltage-Supply(Max):50V Temperature<br>Coefficient:±25ppm/°C Temperature<br>Coefficient:±25ppm/°C Operating<br>Temperature:-55°C~+155°C Operating<br>Temperature:-55°C~+155°C      |  |  |
|  |        |  |  |  |                                                                                                     |  |  |  |                                                                                                                                                                                                                                                                                        |  |  |
|  |        |  |  |  |                                                                                                     |  |  |  | Test Point 测试点                                                                                                                                                                                                                                                                         |  |  |
|  |        |  |  |  |                                                                                                     |  |  |  | CPU Core:ARM Cortex-M Series CPU<br>Maximum Speed:180MHz Program<br>Storage Size:2MB Program Memory<br>Type:FLASH Number of I/O:43 ADC<br>(Bit):12bit                                                                                                                                  |  |  |
|  |        |  |  |  | Enable contr<br>ol ; Over<br>Current Prote<br>ction ; Outpu<br>t discharge ;<br>Thermal<br>shutdown |  |  |  | Output Configuration:Positive Voltage -<br>Supply:5.5V Output Voltage:3.3V Output<br>Voltage:3.3V Output Current:600mA Power<br>Supply Rejection Ratio<br>(PSRR):75dB@(1kHz) Voltage<br>Dropout:320mV@(600mA) standby<br>current:10uA Noise:- Features:Enable<br>control ; Over Curren |  |  |
|  |        |  |  |  |                                                                                                     |  |  |  | Applications Function:USB to serial<br>Protocol:USB 2.0 Number of Channels:-<br>Voltage - Supply:3.3V ; 5V Voltage -<br>Supply:3.3V ; 5V Data Rate:2Mbps<br>Operating Temperature:-20°C~+70°C<br>Operating Temperature:-20°C~+70°C                                                     |  |  |
|  | ±50ppm |  |  |  |                                                                                                     |  |  |  | Frequency:40MHz Voltage - Supply:3.3V<br>Voltage - Supply:3.3V Current -<br>Supply:20mA Frequency Stability:±50ppm<br>Frequency Stability:±50ppm Operating<br>Temperature:-40°C~+85°C Operating<br>Temperature:-40°C~+85°C                                                             |  |  |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-------|--|---|-------|--|-------|--|------|--------------|--|--|
| <a href="https://item.zlcsc.com/datasheet/2011-1X24G00SB/52123955.html">https://item.zlcsc.com/datasheet/2011-1X24G00SB/52123955.html</a>                                                                                             |  |  | 3A    |  |   |       |  |       |  | Gold | Brass        |  |  |
| <a href="https://item.zlcsc.com/datasheet/SM04B-SURS-TF(LF)(SN)/566855.html">https://item.zlcsc.com/datasheet/SM04B-SURS-TF(LF)(SN)/566855.html</a>                                                                                   |  |  | 500mA |  |   |       |  |       |  | Tin  | Copper Alloy |  |  |
| <a href="https://item.zlcsc.com/datasheet/HX%2520PH254-01-02-Z-L11.5%2520male%2520header%2520connector/54760435.html">https://item.zlcsc.com/datasheet/HX%2520PH254-01-02-Z-L11.5%2520male%2520header%2520connector/54760435.html</a> |  |  | 3A    |  |   |       |  |       |  | Gold | Brass        |  |  |
| <a href="https://item.zlcsc.com/datasheet/SCW1005C1N8JST/310235.html">https://item.zlcsc.com/datasheet/SCW1005C1N8JST/310235.html</a>                                                                                                 |  |  | 950mA |  | - |       |  |       |  |      |              |  |  |
| <a href="https://item.zlcsc.com/datasheet/YLED0603G/20555405.html">https://item.zlcsc.com/datasheet/YLED0603G/20555405.html</a>                                                                                                       |  |  |       |  |   |       |  |       |  |      |              |  |  |
| <a href="https://item.zlcsc.com/datasheet/YLED0603R/20382441.html">https://item.zlcsc.com/datasheet/YLED0603R/20382441.html</a>                                                                                                       |  |  |       |  |   |       |  |       |  |      |              |  |  |
| <a href="https://item.zlcsc.com/datasheet/YLED0603R/20382441.html">https://item.zlcsc.com/datasheet/YLED0603R/20382441.html</a>                                                                                                       |  |  |       |  |   |       |  |       |  |      |              |  |  |
| <a href="https://item.zlcsc.com/datasheet/MMBT3904/54682184.html">https://item.zlcsc.com/datasheet/MMBT3904/54682184.html</a>                                                                                                         |  |  |       |  |   | 100nA |  | 200mA |  |      |              |  |  |

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|  |  |  |  | T | SMT |  |  |  |  |  |  |  |  |
|  |  |  |  | B | SMT |  |  |  |  |  |  |  |  |
|  |  |  |  | B | SMT |  |  |  |  |  |  |  |  |
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